

Balaji Rao

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Education

Stevens Institute of Technology, Hoboken, NJ – ABD

(Expected) December 2026

Ph.D., Systems Engineering

Master of Engineering, Engineering Management

May 2022

BNM Institute of Technology, Bengaluru, India

October 2020

Bachelor of Engineering, Electronics and Communication Engineering

Profile

Ph.D. candidate specializing in Systems Engineering with expertise in Large Language Models (LLMs), Neurosymbolic AI, and machine learning solutions. Skilled in building data-driven solutions and structuring LLM outputs into machine-checkable artifacts gated by external AI tools. Demonstrated success in fine-tuning SOTA language models, developing novel algorithms, and optimizing NLP workflows for reliable agentic systems in generative AI.

Skills

Programming Languages: Python, OCaml, Rust, Lean4

Frameworks and Libraries: NumPy, Pandas, PyTorch, TensorFlow, scikit-learn, RAG, GenAI Agents/Tool-Use

Tools and Platforms: Git/GitHub, Claude Code, Kiro, Cursor, AWS (Bedrock, SageMaker AI)

Machine Learning: Natural Language Processing (NLP), Reinforcement Learning, Knowledge Graphs

Experience

Research Assistant - Stevens Institute of Technology

July 2021 - Present

- Mitigated the limitations of probabilistic LLM models by integrating Structured Knowledge, enhancing the generation of coherent and contextually accurate responses at scale.
- Enhanced LLM reasoning for formal verification by developing an automated theorem-proving pipeline in Isabelle/HOL, integrating structured knowledge and reinforcement learning (Pure RL/RLHF/RLVF) to improve accuracy and reliability in safety-critical domains.
- Built LLM-based AI systems by implementing RL training algorithms with fine-tuned accuracy and format rewards, leading to improved coherence, factual accuracy, and reliability in applications.

Applied Science Intern - Automated Reasoning (AWS) (Summer 2025)

May 2025 – August 2025

- Built an LLM-assisted HOL Light/s2n-bignum tool for tactic explanations and tactic suggestions, integrating Bedrock, SageMaker, Kendra, and S3. Interfaced via Amazon Q/VS Code Cline through an MCP server.
- Prototyped a neural theorem-proving agent loop (propose→verify→retry on errors); added evaluation/benchmark artifacts and migrated to a locally hosted model to cut token costs and enable rapid iteration.

Data Science/Data Engineering Intern - Johnson & Johnson (Summer 2024)

May 2024 - August 2024

- Developed machine learning models to analyze and reduce content fatigue, enhancing healthcare professionals' (HCPs) engagement with promotional emails.
- Implemented a Hidden Markov Model (HMM) for probabilistic predictions of email engagement, utilizing a feature matrix that included temporal data. Integrated use of Gen AI solutions to leverage large language models (LLMs) like Llama-2 to optimize content, improving messaging outcomes.
- Introduced new predictive analytics metrics to provide deeper insights into content fatigue, complementing traditional email engagement metrics.

Selected Projects (Publications)

Steve: LLM-Powered ChatBot for Career Progression (AAAI 26)

January 2026

- Built an ontology-driven RAG stack with hybrid retrieval and schema-validated function calling to compute skill-gaps and personalized upskilling paths. Implemented agentic AI loops for resume parsing, skills mapping, and course selection; added guardrails. Productionized leveraging LLM tool-use for deterministic, auditable flows.

SSE-EduBot: Course-Specific LLM Tutoring Assistant (FIE 2025)

November 2025

- Built a syllabus-aligned RAG tutor and fine-tuned Llama-2-13B (LoRA/PEFT) for domain-specific answers; benchmarked against ChatGPT-3.5 with more context-aligned explanations and fewer domain-specific hallucinations; deployed on a university server to collect interactions and refine retrieval precision, latency, and guardrails.

Neural Theorem Proving for Formal Verification (NeSy 2025)

September 2025

- Built ProofSeek, a two-stage (SFT + RL/GRPO) LLM framework for whole-proof generation in Isabelle/HOL with autoformalization and ProofAug-based verification; on an LLM-generated AWS S3 policy benchmark, improved success by 3% and reduced runtime by 20%, while matching curated-policy accuracy with faster runs.